Math 105L In Class 2 Practice

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Derivatives of logs

1. Differentiate $y = \ln(x^2 + 1)$.

2. Differentiate $y = \log(3x)$.

3. Differentiate $y = \ln\left(1 + e^{2x}\right)$.

4. Differentiate $y = (\ln x)^2$ for $x > 0$.

5. Differentiate $y = \ln\left(\frac{x^2}{\sqrt{1+x}}\right)$ for $x > 0$.

6. Differentiate $y = \ln\left(\sqrt{x^2 + 4}\right)$.

7. Differentiate $y = \ln(x) \cdot \ln(2x)$ for $x > 0$.

8. Differentiate $y = \ln\left(x + \sqrt{1 + x^2}\right)$.

9. Differentiate $y = \log(x^3 + 1)$.

10. Differentiate $y = \ln\big(f(x)\big)$ where $f(x) = x^2 e^{3x}(x + 1)$ (assume x where $f(x) > 0$).

Using First and Second Derivatives

1. For each of the following, find and classify all critical points. Find all inflection points too.

   (a) $f(x) = x^3 - 3x^2 - 9x + 27 = (x-3)^2(x+3)$

   (b) $f(x) = x^4 - 4x^3 + 4x^2 = x^2(x-2)^2$

   (c) $f(x) = x^3 - 3x^2 + 3x - 1 = (x-1)^3$

   (d) $f(x) = x^4 - 2x^2 = x^2(x-\sqrt{2})(x+\sqrt{2})$

   (e) $f(x) = x^3 - 3x = x(x-\sqrt{3})(x+\sqrt{3})$

   (f) $f(x) = xe^x - e^x = e^x(x-1)$

2. Give examples of functions satisfying each of the following sets of conditions:

   (a) $f'(0) = 0$, but $f'(x)$ does not have a max or min at $x = 0$

   (b) $f'(0) = 0$, $f''(0) = 0$, and $f'(x)$ has a max at $x = 0$.

   (c) $f''(x) = 0$, but $f(x)$ does not have an IP at $x = 0$.

   (d) $f(x)$ has a min at $x = 0$, but $f'(0) \neq 0$.

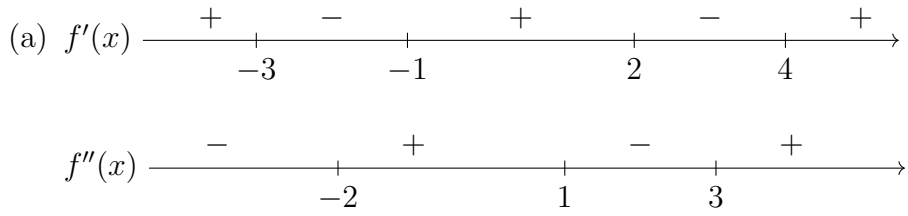
   (e) $f''(0) > 0$, but $f(x)$ does not have a min at $x = 0$.

   (f) $f'(0)$ is defined at $x = 0$, $f''(0)$ is not defined, and $f(x)$ has an IP at $x = 0$ (This last one is hard – skip it if you want. It will not be on the test.)

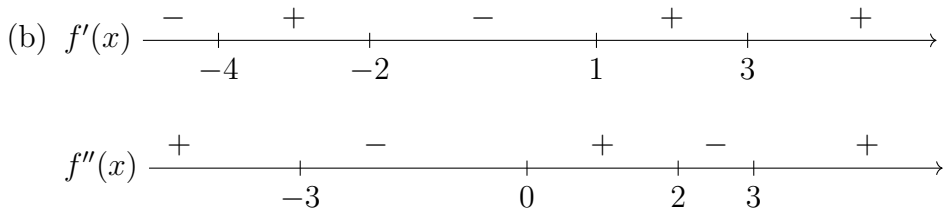
3. In each part below, number lines are presented for the first and second derivatives of a function. Find and classify all critical points. Find all inflection points.

   Also: precisely one of these has an error where the classification is logically inconsistent. Find it and explain.

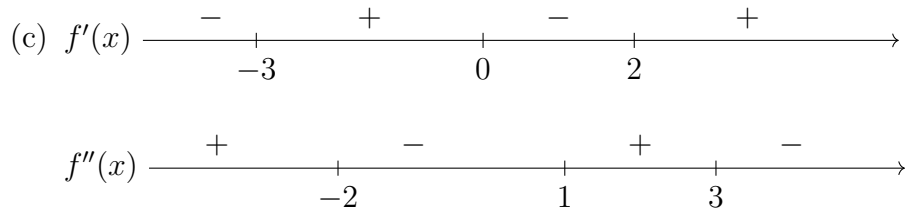
(a)



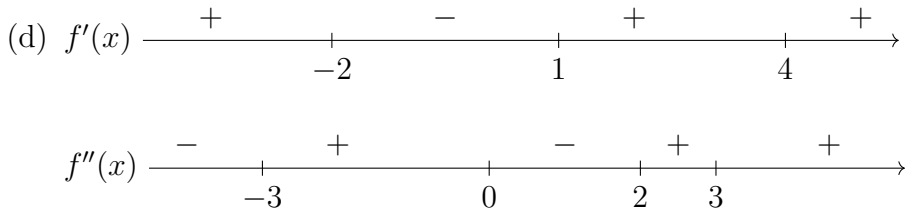
(b)



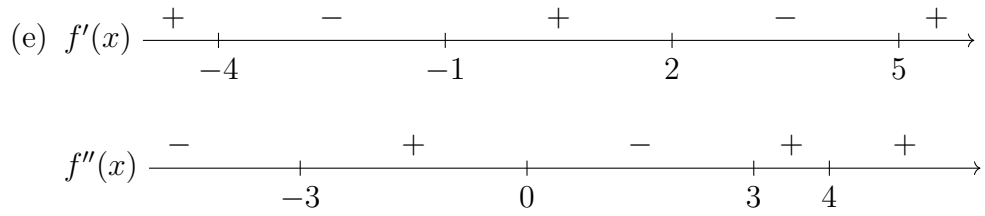
(c)



(d)



(e)



Global Max's and Min's

For each of the following functions, find its global maximum and/or global minimum on the given domain, if they exist. Pay careful attention to cases where the domain is not closed or not finite!

1. $f(x) = \dfrac{1}{x}$ for $0 < x < 1$.
2. $f(x) = x^3 - 3x$ for $-2 < x < 2$.
3. $f(x) = x^3 - 3x^2 + 1$ for $0 < x < 4$.
4. $f(x) = x^2$ for $-3 \leq x \leq 3$.
5. $f(x) = -x^2 + 4x - 1$ for $0 \leq x \leq 4$.
6. $f(x) = e^{-x}$ for $0 \leq x < \infty$.
7. $f(x) = \ln(x)$ for $0 < x \leq 5$.
8. $f(x) = xe^{-x}$ for $0 < x < \infty$.
9. $f(x) = x^2e^{-x}$ for $0 \leq x < \infty$.
10. $f(x) = \ln(x^2 + 1)$ for $-3 < x < 3$.

Optimization

1. **Open-Top Wooden Storage Chest**

   A carpenter is building an *open-top storage chest* with a *rectangular base* whose *length is twice its width*. The chest must have a volume of 120 cubic inches. The wood for the base costs 5¢ per square inch, and the wood for the sides costs 2¢ per square inch.

   Find: The dimensions of the chest that minimize the total cost of materials.

2. **Open-Top Cylindrical Can**

   A manufacturer wants to design an open-top cylindrical can that will hold 500 cm^3 of liquid. The material for the sides costs 2¢ per cm^2, and the material for the base costs 5¢ per cm^2.

   Find: The radius and height of the can that minimize the total cost.

3. **Fencing Along a River**

   A rectangular field is to be fenced along a straight river. The side along the river requires no fence, and you have 100 m of fencing for the remaining three sides.

   Find: The dimensions that maximize the enclosed area.

4. **Two Numbers with Fixed Sum**

   Find two positive numbers whose sum is 12 and whose product is as large as possible.

5. **Closest Point on a Parabola**

   Find the point $Q(x, y)$ on the parabola $y = x^2$ that is closest to the point $P = (0, 3)$.

   (Hint: The coordinates of a point on a parabola are (x, x^2). Find a formula for the square of the distance between P and this point, then minimize it.)

Linear Approximations

1. Approximate the value of $\sqrt[3]{27.1}$ using a linear approximation near $x = 27$.
   Without computing the exact value of $\sqrt[3]{27.1}$, is your estimate an overestimate or underestimate? Explain your answer.

2. Use linear approximation to estimate the value of $\sqrt{49.2}$ by approximating the function $f(x) = \sqrt{x}$ near $x = 49$.
   Without computing the exact value of $\sqrt{49.2}$, is your estimate an overestimate or underestimate? Explain your answer.

3. Use the linear approximation to estimate $f(1.1)$ where $f(x) = x^3 - 2x + 5$. Approximate near $x = 1$.
   Without computing the exact value of $f(1.1)$, is your estimate an overestimate or underestimate? Explain your answer.

4. Approximate the value of $\ln(1.05)$ using a linear approximation near $x = 1$.
   Without computing the exact value of $\ln(1.05)$, is your estimate an overestimate or underestimate? Explain your answer.

5. Use a linear approximation to estimate $\frac{1}{2.1}$ by approximating the function $f(x) = \frac{1}{x}$ near $x = 2$.
   Without computing the exact value of $\frac{1}{2.1}$, is your estimate an overestimate or underestimate? Explain your answer.

6. Use the linear approximation to estimate $f(0.9)$ where $f(x) = x^3 + 3x - 2$. Approximate near $x = 1$.
   Without computing the exact value of $f(0.9)$, is your estimate an overestimate or underestimate? Explain your answer.

7. Use the linear approximation to estimate $f(1.1)$ where $f(x) = e^x$. Approximate near $x = 1$.
   Without computing the exact value of $f(1.1)$, is your estimate an overestimate or underestimate? Explain your answer.

L'Hopital's Rule

Compute the following limits. Check at each step if L'Hôpital is applicable. Show all your work.

1. $\lim_{x\to 0} \dfrac{e^x - 1 - x}{x^2}$
2. $\lim_{x\to 0} \dfrac{e^{2x} - 1 - 2x}{x}$
3. $\lim_{x\to \infty} \dfrac{x}{e^x}$
4. $\lim_{x\to 0^+} x \ln x$
5. $\lim_{x\to 0} \dfrac{1 - e^{-x}}{x}$
6. $\lim_{x\to 0} \dfrac{\ln(1+x) - x}{x^2}$
7. $\lim_{x\to 0} \dfrac{\ln(1+x) - x + \frac{x^2}{2}}{x^3}$
8. $\lim_{x\to 0} \dfrac{e^x - 1}{x}$
9. $\lim_{x\to \infty} \dfrac{\ln x}{x}$
10. $\lim_{x\to 0} \dfrac{\sqrt{1+x} - 1 - \frac{x}{2}}{x^2}$
11. $\lim_{x\to 0} \dfrac{1 - \sqrt{1-x}}{x}$
12. $\lim_{x\to 0} \dfrac{e^x - 1 - x - \frac{x^2}{2}}{x^3}$
13. $\lim_{x\to \infty} \dfrac{e^x}{x^2}$
14. $\lim_{x\to 0} \dfrac{e^x + 1}{x}$
15. $\lim_{x\to 0} \dfrac{e^x - 1 - x - \frac{x^2}{2} - \frac{x^3}{6}}{x^4}$

Related Rates

1. **Inflating a Spherical Balloon**
   Air is pumped into a spherical balloon so that its surface area increases at 24 cm^2/s.
   (a) How fast is the radius increasing when the radius is 3 cm?
   (b) How fast is its volume changing at that same instant?

2. **Water Filling a Conical Tank**
   Water fills a conical tank.

   (a) Explain why, if the radius is 2 ft when the water depth is 6 ft, it is always true that $r = \frac{1}{3}h$.

   (b) The volume increases at 2 ft^3/min. How fast is the height rising when $h = 6$ ft?

3. **Expanding Circular Puddle**
   A circular puddle's area is increasing at 10 m^2/hr. When the radius is 5 m, how fast is the radius increasing?

4. **Rectangle With One Side Fixed**
   A poster has height 12 in and width increasing at 0.5 in/s. How fast is its area increasing when the width is 8 in?

5. **Growing Cylinder**
   A Cylinder has radius 2 m at a certain instant. Its radius is growing at 0.03 m/min. The water depth is always 3 m. How fast is its volume changing?

6. **Two Cars**
   Car A moves east at 60 mph and Car B north at 40 mph. At one moment, A is 3 miles east and B is 4 miles north of the intersection. How fast is the distance between them changing?

7. **Inflating Cube**
   A cube's surface area increases at 30 cm^2/s. How fast is the edge increasing, and how fast is the volume increasing, when the edge is 5 cm?

8. **Rectangle with Fixed Diagonal**
   A rectangle has diagonal 13 cm. Its width increases at 0.2 cm/s. How fast is the area changing when the width is 5 cm?

   (Hint: relate the width and height to the length of the diagonal)